

College Football Chaos

Elo predicts strength. Chaos predicts instability.

League-Wide Chaos

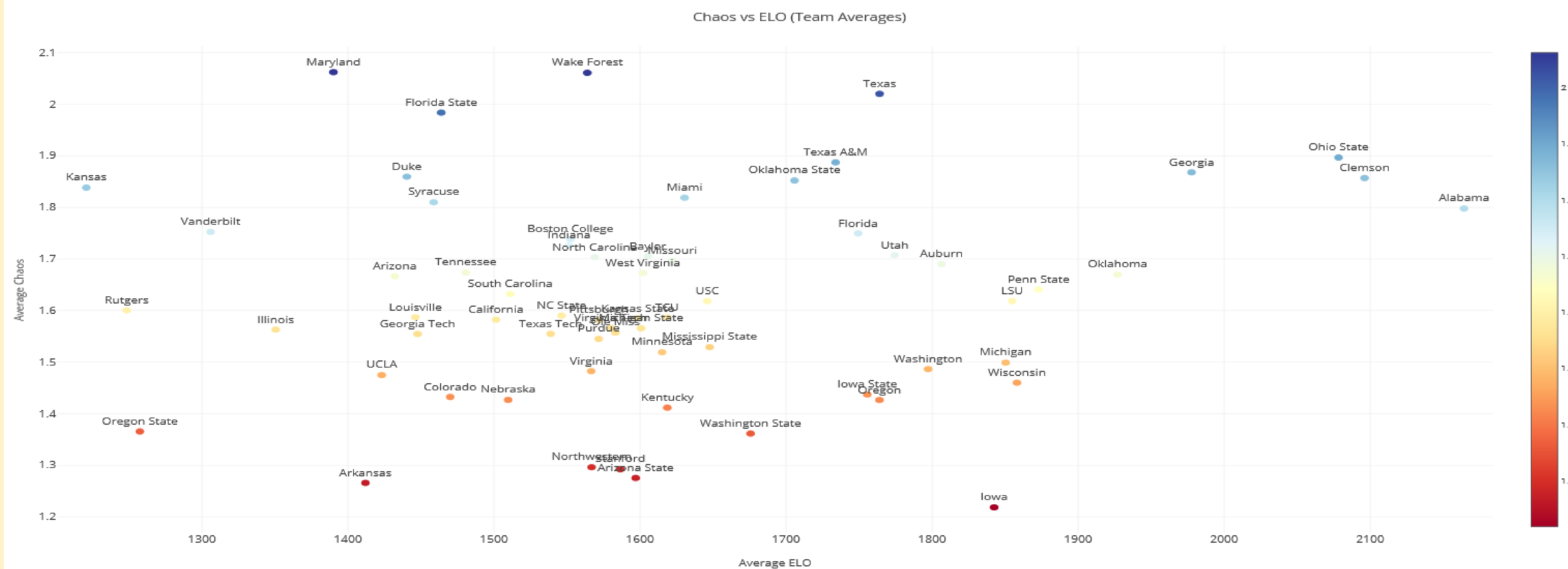


Fig 1. Teams cluster into stable vs volatile profiles

Chaos by Team

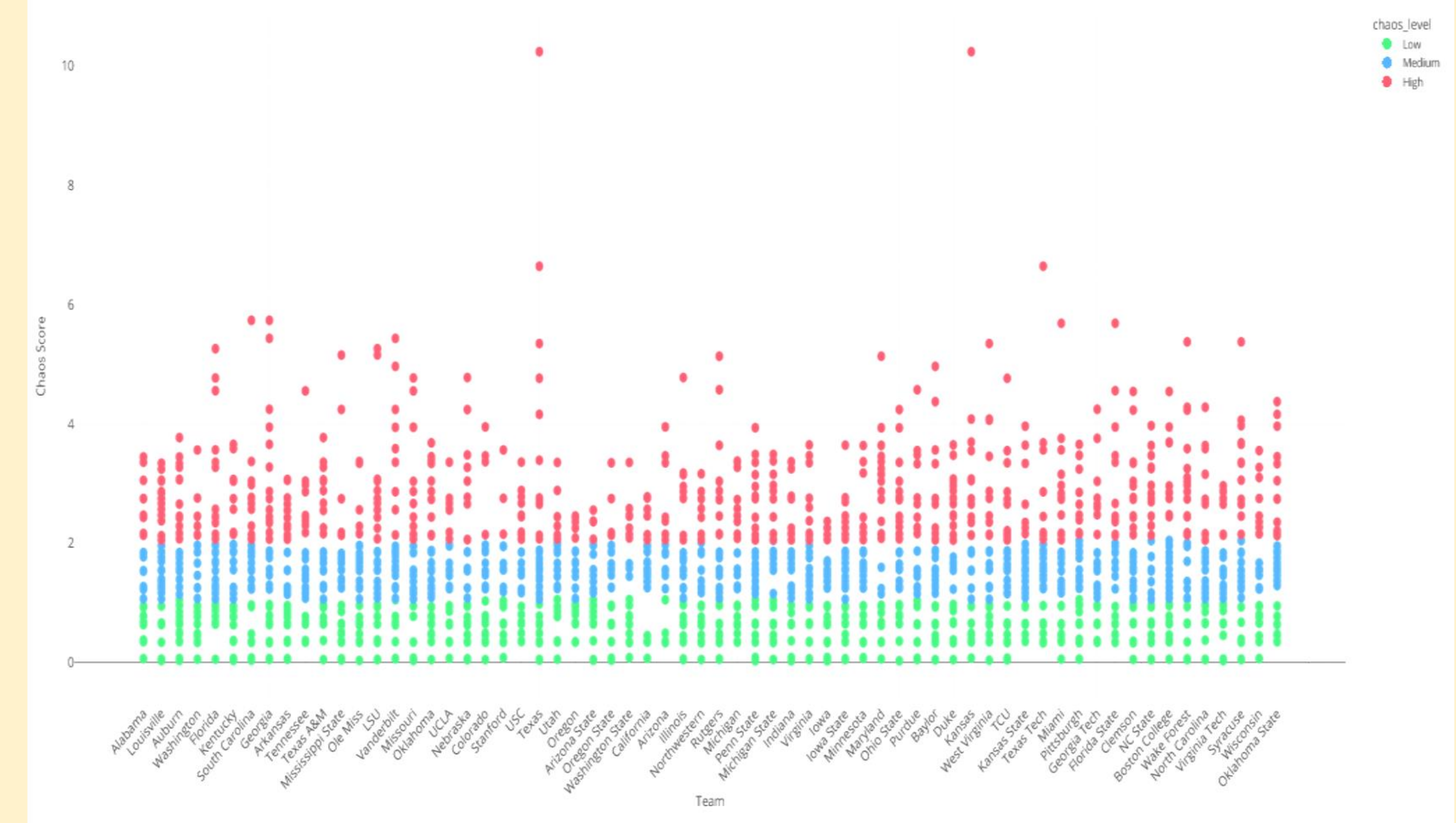


Fig 2. Chaos highlights structural tendencies not visible in Elo.

THE CHAOS FACTOR

$$Chaos = W_1 \cdot WinProbabilityVolatility + W_2 \cdot LeadChanges + W_3 \cdot ExplosivePlaysDelta$$

Why Chaos?

- Elo measures strength -- not volatility
- Momentum swings break predictions
- Explosive plays create instability
- No metric captures true *game chaos*
- We introduce the **Chaos Factor**

Weekly Chaos v Elo Gap

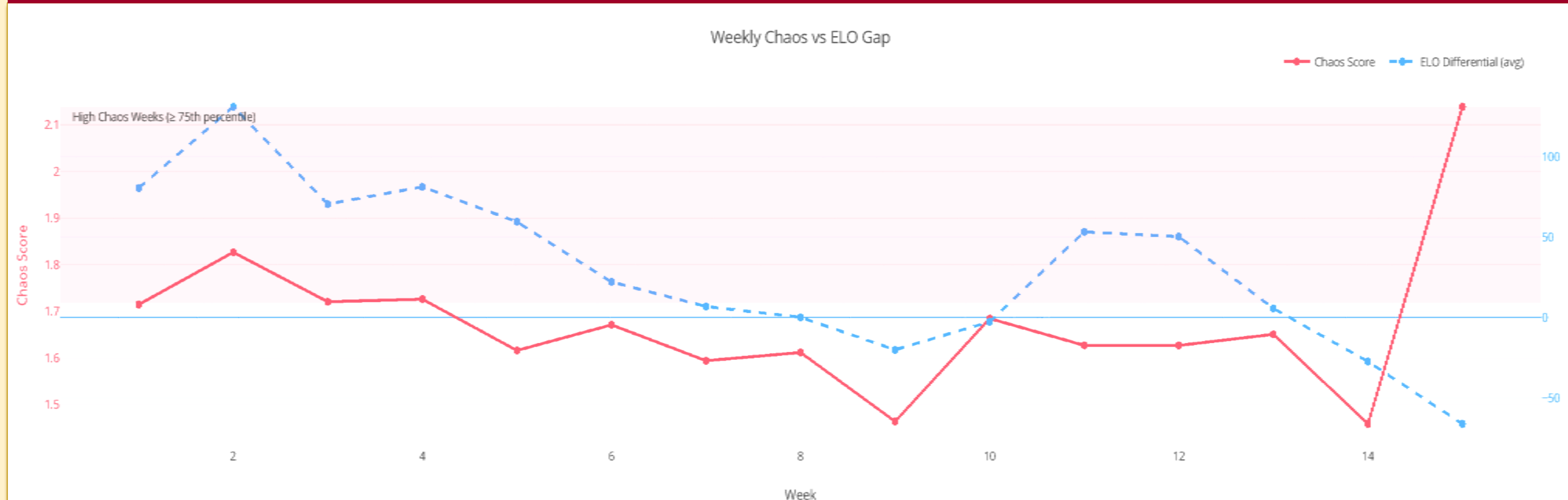


Fig. 3 Chaos spikes coincide with rivalry and playoff games
Elo misses instability signals

Rivalries Defy Elo

Matchup	Avg Chaos
Kansas vs Texas	4.41
Georgia vs Vanderbilt	3.27
Florida vs Florida St	2.95
Auburn vs Texas A&M	2.91
Maryland vs Rutgers	2.89

Key Takeaway

Strength ≠ Stability

Chaos exposes hidden instability that Elo misses

